

# Semiparametric Difference-in-Differences Estimators

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The difference-in-differences (DID) estimator is one of the most popular tools for applied research in economics to evaluate the effects of public interventions and other treatments of interest on some relevant outcome variables. However, it is well known that the DID estimator is based on strong identifying assumptions. In particular, the conventional DID estimator requires that, in the absence of the treatment, the average outcomes for the treated and control groups would have followed parallel paths over time. This assumption may be implausible if pre-treatment characteristics that are thought to be associated with the dynamics of the outcome variable are unbalanced between the treated and the untreated. That would be the case, for example, if selection for treatment is influenced by individual-transitory shocks on past outcomes (Ashenfelter's dip). This article considers the case in which differences in observed characteristics create non-parallel outcome dynamics between treated and controls. It is shown that, in such a case, a simple two-step strategy can be used to estimate the average effect of the treatment for the treated. In addition, the estimation framework proposed in this article allows the use of covariates to describe how the average effect of the treatment varies with changes in observed characteristics.

A good way to do econometrics is to look for good natural experiments *and* use statistical methods that can tidy up the confounding factors that nature has not controlled for us.  
(Daniel McFadden, *Econometric Tools*)

## 1. INTRODUCTION

The use of natural experiments to evaluate treatment effects in the absence of truly experimental data has gained wide acceptance in empirical research in economics and other social sciences. Simple comparisons of pre-treatment and post-treatment outcomes for those individuals exposed to a treatment are likely to be contaminated by temporal trends in the outcome variable or by the effect of events, other than the treatment, that occurred between both periods. However, when only a fraction of the population is exposed to the treatment, an untreated comparison group can be used to identify temporal variation in the outcome that is not due to treatment exposure. The difference-in-differences (DID) estimator is based on this simple idea. Card and Krueger (1994) assess the employment effects of a raise in the minimum wage in New Jersey using a neighbouring state, Pennsylvania, to identify the variation in employment that New Jersey would have experienced in the absence of a raise in the minimum wage. Other applications of DID include studies of the effects of immigration on native wages and employment (Card, 1990), the effects of temporary disability benefits on time out of work after an injury (Meyer, Viscusi and Durbin, 1995), and the effect of anti-takeover laws on firms' leverage (Garvey and Hanka, 1999).

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average outcomes for treated and controls would have followed parallel paths over time. This assumption may be implausible if pre-treatment characteristics that are thought to be associated with the dynamics of the outcome variable are unbalanced between the treated and the untreated group.

This study considers the case in which differences in observed characteristics create non-parallel outcome dynamics for the treated and untreated groups. It is shown that, in such a case, a simple two-step strategy can be used to estimate the average effect of the treatment for the treated. In addition, the estimation framework proposed in this article allows the use of covariates to describe how the average effect of the treatment varies with changes in observed characteristics.

Despite the prolific literature on semiparametric and non-parametric methods, few articles have been devoted to studying and relaxing the DID identification restrictions. Some exceptions are Besley and Case (1994), Meyer (1995), Heckman, Ichimura and Todd (1997), Imbens, Liebman and Eissa (1997), Heckman, Ichimura, Smith and Todd (1998), Angrist and Krueger (1999), Blundell and MaCurdy (1999), Blundell, Costa Dias, Meghir and van Reenen (2001) and Athey and Imbens (2002).

The identification procedure used in this article originated with Heckman *et al.* (1997, 1998). However, the estimation procedure differs from the earlier literature in three ways. First, it does not require repeated observations for the same individuals. The proposed estimators are feasible under the data requirements for traditional DID estimators when applied to repeated cross-sections. Second, it allows the estimation of parsimonious parametric approximations to the average effect of the treatment on the treated conditional on selected covariates of interest. Finally, the framework can accommodate multilevel treatment variables (that is, different treatment intensities).

The rest of the article is organized as follows. Section 2 describes the conventional DID model and discusses some of its limitations. Section 3 presents the main identification results of the article, followed by some extensions. The estimation strategy along with asymptotic distribution theory is provided in Section 4. Section 5 concludes. Proofs are presented in the Appendix.

## 2. THE DID ESTIMATOR

The basic DID framework can be described as follows. Let  $Y(i, t)$  be the outcome of interest for individual  $i$  at time  $t$ . The population is observed in a pre-treatment period  $t = 0$ , and in a post-treatment period  $t = 1$ . Between these two periods, some fraction of the population is exposed to the treatment. We denote  $D(i, t) = 1$  if individual  $i$  has been exposed to the treatment previous to period  $t$ ,  $D(i, t) = 0$  otherwise. We call those individuals with  $D(i, 1) = 1$  *treated*, and those with  $D(i, 1) = 0$  *controls* (or *untreated*). Since individuals are only exposed to treatment after the first period,  $D(i, 0) = 0$  for all  $i$ .

The conventional DID estimator is often derived using a linear parametric model. It is useful to consider this formulation of the DID model first, to fix ideas, before studying non-parametric identification in Section 3. The following formulation of the DID model is based on that given in Ashenfelter and Card (1985). Suppose that the outcome variable is generated by a components of variance process

$$Y(i, t) = \delta(t) + \alpha \cdot D(i, t) + \eta(i) + v(i, t), \quad (1)$$

where  $\delta(t)$  is a time-specific component,  $\alpha$  represents the impact of the treatment,  $\eta(i)$  is an individual-specific component, and  $v(i, t)$  is an individual-transitory shock that has mean zero at each period,  $t = 0, 1$ , and is possibly correlated in time. Only  $Y(i, t)$  and  $D(i, t)$  are observed. The effect of the treatment,  $\alpha$ , is not identified without further restrictions. A sufficient condition

for identification is that selection for treatment does not depend on the individual-transitory shocks, that is

$$P(D(i, 1) = 1 \mid v(i, t)) = P(D(i, 1) = 1) \quad (2)$$

for  $t = 0, 1$ . Adding and subtracting  $E[\eta(i) \mid D(i, 1)]$  in equation (1), we obtain

$$Y(i, t) = \delta(t) + \alpha \cdot D(i, t) + E[\eta(i) \mid D(i, 1)] + \varepsilon(i, t), \quad (3)$$

where  $\varepsilon(i, t) = \eta(i) - E[\eta(i) \mid D(i, 1)] + v(i, t)$ . Notice that  $\delta(t) = \delta(0) + (\delta(1) - \delta(0))t$ , and  $E[\eta(i) \mid D(i, 1)] = E[\eta(i) \mid D(i, 1) = 0] + (E[\eta(i) \mid D(i, 1) = 1] - E[\eta(i) \mid D(i, 1) = 0])D(i, 1)$ . Let  $\mu = E[\eta(i) \mid D(i, 1) = 0] + \delta(0)$ ,  $\tau = E[\eta(i) \mid D(i, 1) = 1] - E[\eta(i) \mid D(i, 1) = 0]$  and  $\delta = \delta(1) - \delta(0)$ . We obtain

$$Y(i, t) = \mu + \tau \cdot D(i, 1) + \delta \cdot t + \alpha \cdot D(i, t) + \varepsilon(i, t). \quad (4)$$

The restriction in equation (2) for  $t = 0, 1$  implies  $E[(1, D(i, 1), t, D(i, t)) \cdot \varepsilon(i, t)] = 0$ , so all the parameters in equation (4), including the treatment impact  $\alpha$ , are estimable by least squares. Notice that the model allows *any* kind of dependence between selection for treatment,  $D(i, 1) = 1$ , and the individual-specific component,  $\eta(i)$ . This model is called “difference-in-differences” because under the identifying condition in equation (2) we have

$$\begin{aligned} \alpha &= \{E[Y(i, 1) \mid D(i, 1) = 1] - E[Y(i, 1) \mid D(i, 1) = 0]\} \\ &\quad - \{E[Y(i, 0) \mid D(i, 1) = 1] - E[Y(i, 0) \mid D(i, 1) = 0]\}, \end{aligned} \quad (5)$$

and the least squares estimator of  $\alpha$  is the sample counterpart of equation (5).

This formulation of the problem is useful when repeated cross sections of  $(Y(i, t), D(i, 1))$  for  $t = 0, 1$  are available. If a sample with repeated pre-treatment and post-treatment observations of the outcome variable,  $Y(i, 1)$  and  $Y(i, 0)$ , is available, then  $\alpha$  is estimable by least squares regression of  $Y(i, 1) - Y(i, 0)$  on  $D(i, 1)$ :

$$\alpha = E[Y(i, 1) - Y(i, 0) \mid D(i, 1) = 1] - E[Y(i, 1) - Y(i, 0) \mid D(i, 1) = 0].$$

Note that equation (2) for  $t = 0, 1$  implies that  $v(i, 1) - v(i, 0)$  is mean independent of  $D(i, 1)$ , and therefore that, in absence of the treatment, the average outcome for the treated would have experienced the same variation as the average outcome for the untreated. This restriction, implied by the model, may be too stringent if treated and controls are unbalanced in covariates that are thought to be associated with the dynamics of the outcome variable.

For example, it has been documented that participants in training programmes experience a decline in earnings prior to the training period (Ashenfelter’s dip, Ashenfelter (1978)). This fact suggests that selection for training may be affected by individual-transitory shocks in pre-training earnings. To accommodate this conjecture, Ashenfelter and Card (1985) propose the following model for the selection process:

$$D(i, 1) = \begin{cases} 1 & \text{if } Y(i, 1 - \kappa) + u(i) < \bar{Y} \\ 0 & \text{otherwise,} \end{cases} \quad (6)$$

where  $\kappa$  is a positive integer,  $\bar{Y}$  is a constant, and  $u(i)$  is a random variable independent of any variance component. Under this formulation for the selection process, those individuals with low earning  $\kappa$  periods before training are more likely to participate in the training programme. The identification condition in equation (2) does not hold in general for this example. The reason is that the individual-specific components,  $v(i, t)$ , are allowed to be correlated in time. However, if the selection process can be represented by equation (6), then  $P(D(i, 1) = 1 \mid Y(i, 1 - \kappa), v(i, t)) = P(D(i, 1) = 1 \mid Y(i, 1 - \kappa))$ . The DID model holds conditional on

$Y(i, 1 - \kappa)$ , so the impact of the treatment is given by

$$\begin{aligned} & \{E[Y(i, 1) | X(i), D(i, 1) = 1] - E[Y(i, 1) | X(i), D(i, 1) = 0]\} \\ & - \{E[Y(i, 0) | X(i), D(i, 1) = 1] - E[Y(i, 0) | X(i), D(i, 1) = 0]\} \end{aligned} \quad (7)$$

where  $X(i) = Y(i, 1 - \kappa)$ .<sup>1</sup> More generally, in this article  $X(i)$  is a vector of observed characteristics, such as demographic attributes, predetermined at  $t = 0$ . A conditional identification restriction is appealing in the DID framework when the variables in  $X(i)$  are believed to be related to the outcome dynamics and their distributions differ between treated and controls.

The traditional way to accommodate covariates in the DID model is to introduce them linearly in equation (4):

$$Y(i, t) = \mu + X(i)' \pi(t) + \tau \cdot D(i, 1) + \delta \cdot t + \alpha \cdot D(i, t) + \varepsilon(i, t), \quad (8)$$

where  $X(i)$  is assumed uncorrelated with  $\varepsilon(i, t)$ . Because the coefficients on  $X(i)$  change with  $t$ , this formulation of the DID model allows the use of covariates to represent heterogeneity in outcome dynamics. Differencing the last equation with respect to  $t$ , we obtain

$$Y(i, 1) - Y(i, 0) = \delta + X(i)' \pi + \alpha \cdot D(i, 1) + (\varepsilon(i, 1) - \varepsilon(i, 0)),$$

where  $\pi = \pi(1) - \pi(0)$ . This alternative formulation is useful when a sample with repeated observations is available. However, as noticed by Meyer (1995), introducing covariates in this linear fashion may not be appropriate if the treatment has different effects for different groups in the population. Heterogeneity in treatment effects can be studied by specifying  $\alpha$  in equation (8) as a function of  $X(i)$  (i.e. by including interactions between  $X(i)$  and  $D(i, t)$  in equation (8)). Ideally, covariates should be treated non-parametrically, as in equation (7), so that any potential inconsistency created by functional form misspecification is avoided. However, when the number of covariates required to attain identification is large, some kind of integration over  $X(i)$  is required in order to obtain interpretable results.

The next section proposes a flexible new procedure to control for the effect of covariates in the DID model which is based on conditional identification restrictions. As in the conventional DID model, the role of the covariates in this new approach is twofold. First, by using covariates we extend identification to those instances in which observed compositional differences between treated and controls cause non-parallel dynamics in the outcome variable. In addition, the effect of the treatment is allowed to differ among individuals, so covariates can be used to describe the effect of the treatment for different groups of the population. A distinctive feature of the methods proposed in this article is that, while covariates are treated non-parametrically for identification, the estimators provide parsimonious parametric approximations to the average effect of the treatment on the treated conditional on selected covariates of interest.<sup>2</sup>

A related way to accommodate covariates in a DID estimator has been explored by Heckman *et al.* (1997, 1998) who propose a DID estimator of the average treatment effect on the treated based also on conditional identification restrictions. Their estimator is constructed by matching differences in pre-treatment and post-treatment outcomes for the treated to weighted averages of differences in pre-treatment and post-treatment outcomes for the untreated. The differences are matched on the probability of treatment exposure conditional on the covariates

1. In fact, for Ashenfelter's dip model in equation (6), the second term in equation (7) is zero and estimation could be based solely on conditional averages of post-treatment outcomes.

2. In other words, while identification does not hinge on parametric restrictions, it allows the use of parametric functions to describe how the effect of the treatment varies with covariates. This approach follows the spirit of White (1981) and Roehrig (1988) among others who propose treating parametric models as convenient approximations to unknown functions of interest, so identification can be studied non-parametrically.